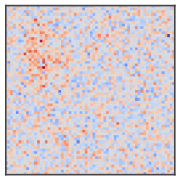
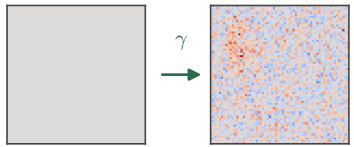
Inputs

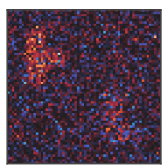


$\mathbf{x}_\star$ target instance

f differentiable classifier

1 Deterministic IG



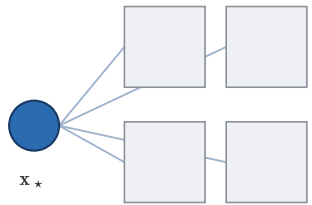


$\mathbf{w}^{\mathrm{PG}} = \hat{\mathbf{s}}^{\mathrm{IG}}(\mathbf{x}_\star)$

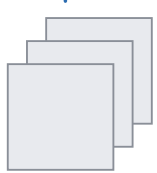
pixel-level; M-step quadrature;
no random masks, no surrogate fit;
deterministic for fixed $(f, \mathbf{x}', M)$

2 Neighborhood

$\mathcal{N}_K(\mathbf{x}_\star)$

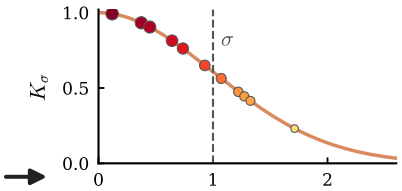


K nearest in $\mathcal{D}_{\mathrm{train}}$
(Euclidean); no perturbations



$\mathcal{D}_{\mathrm{train}}$: the K nearest neighbors
of $\mathbf{x}_\star$ are retrieved here

3 Gaussian locality



$\pi_j = K_\sigma(d_j) \,/\, \sum_{\ell \in \mathcal{N}_K} K_\sigma(d_\ell)$

$d_j = \|\mathbf{x}^{(j)} - \mathbf{x}_\star\|_2$ (units of σ)



$\widetilde{\mathbf{X}}_K$: $\tilde{\mathbf{x}}^{(j)} = \mathbf{x}^{(j)} - \boldsymbol{\mu}$, $\boldsymbol{\mu} = \sum_j \pi_j \mathbf{x}^{(j)}$

adaptive, hyper-parameter-free;
not a retraining step

4 Conversion

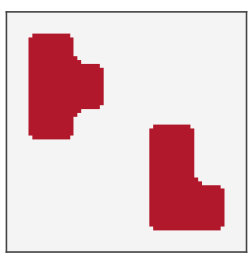
$$\mathbf{a} = \frac{\mathbf{C}_K \mathbf{w}^{\mathrm{PG}}}{(\mathbf{w}^{\mathrm{PG}})^\top \mathbf{C}_K \mathbf{w}^{\mathrm{PG}} + \lambda}$$

$\mathbf{C}_K \mathbf{w}^{\mathrm{PG}} = \mathrm{Cov}_{\Pi_K(\mathbf{x}_\star)}[\mathbf{x}, \tilde{y}]$

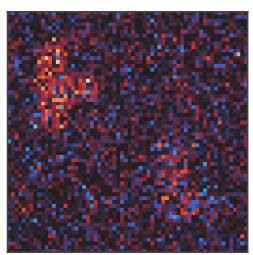
$\mathbf{w}^{\mathrm{PG}}$ → never formed → $\mathbf{a}$

$\widetilde{\mathbf{X}}_K^\top[\boldsymbol{\pi} \odot (\widetilde{\mathbf{X}}_K \mathbf{w}^{\mathrm{PG}})]$

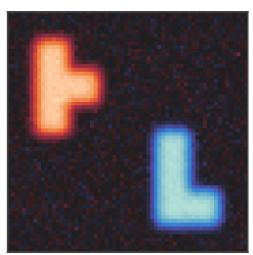
$\mathcal{O}(KD)$: covariance conversion only



$\mathcal{F}^+$ (ground truth)



raw Integrated Gradients



PatternGrad

Legend

input data; binary support $\mathcal{F}^+$

attributions: signed, max-$|\cdot|$ normalized,
identical signed-sqrt display

Held-out example: LIN-WHITE, $\alpha = 0.2$, $y = 0$,
test index 78 (first held-out eval sample)

PatternGrad: deterministic pixel-level IG + covariance conversion on the training neighborhood.
PatternLocal: stochastic binary-mask perturbations, per-instance surrogate refit.